An Introduction to Digital System Design using SystemVerilog and FPGA

Lecture 1, Fall 2026

Digital System Design EE 421 / CS 425

Shahid Masud

Important Goals of this Course

- Design Complex and High-Speed Digital Systems
- Understand Timing Problems in Complex High-Speed Digital Design
- Analyze and Synthesize Complex Digital Systems (combinational and sequential)
- Learn some techniques to develop High-Speed Complex Digital Systems using modern tools (~~Modelsim~~, Questa, Xilinx) and modern implementation platforms (FPGA)
- Introducing SystemVerilog and Verification

LABS

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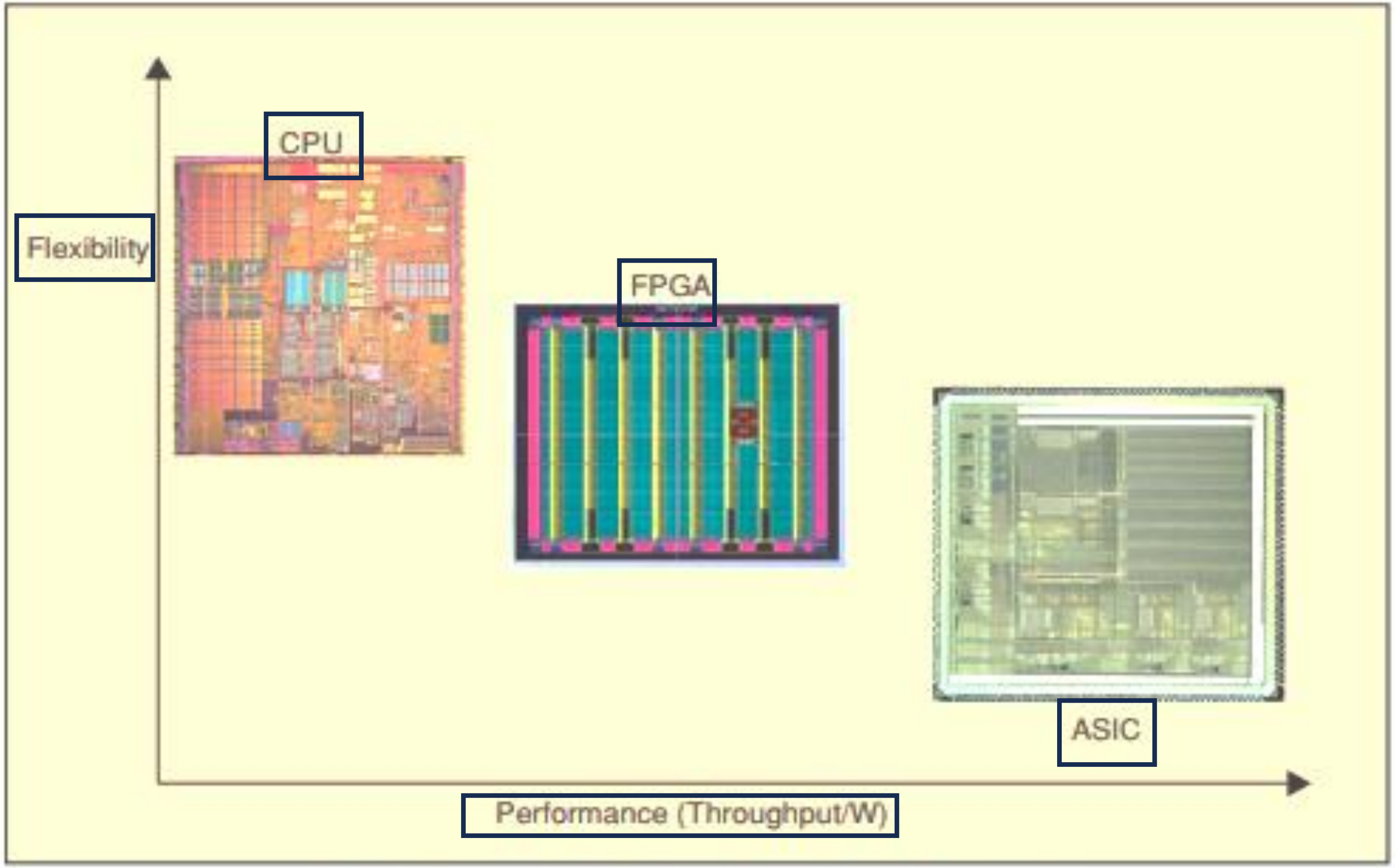
Previous Courses

Custom PCB Design	Microcontroller/ PLC Based Design	FPGA Programmable Logic Design	ASIC full custom Chip design
Small scale, sub-system level, glue logic, limited functionality as per available chips, specialized skill for PCB design at high-speeds.	Functionality can be upgraded through software, flexible but slow speed. More peripherals can be added or removed.	Faster than microcontroller. Flexible but requires high-level HDL and EDA tools knowledge. Quick design and prototyping.	Very fast and high performance. Design and fabrication time is significant. Produced in large numbers. Rigid functionality. HDL and EDA tools needed.

Implementation Platforms



1. Flexibility versus performance for CPU, FPGA, and ASIC.

Ref: IEEE Circuits and Devices Magazine, May 2006

Examples of Custom Digital Systems

Special purpose circuits and chips

Cybersecurity and Chips

China to probe Micron over cybersecurity, in chip war's latest battle

The Chinese government will investigate US-based Micron as a potential cyberthreat, in the latest move in an ongoing semiconductor trade dispute that is disrupting the chip supply chain.

By Jon Gold
Senior Writer, Network World | APR 3, 2023 2:15 PM PDT

The Chinese government is instituting a cybersecurity review of US-based memory chip maker Micron's products being sold in the country, in the latest move in the ongoing semiconductor trade dispute that pits China against the US and its allies.

https://www.channelnews.com.au/new-faster-processing-chip-unveiled-by-nvidia/

NEW FASTER PROCESSING CHIP UNVEILED BY NVIDIA

By Ashley Riordan| 9 Aug 2023

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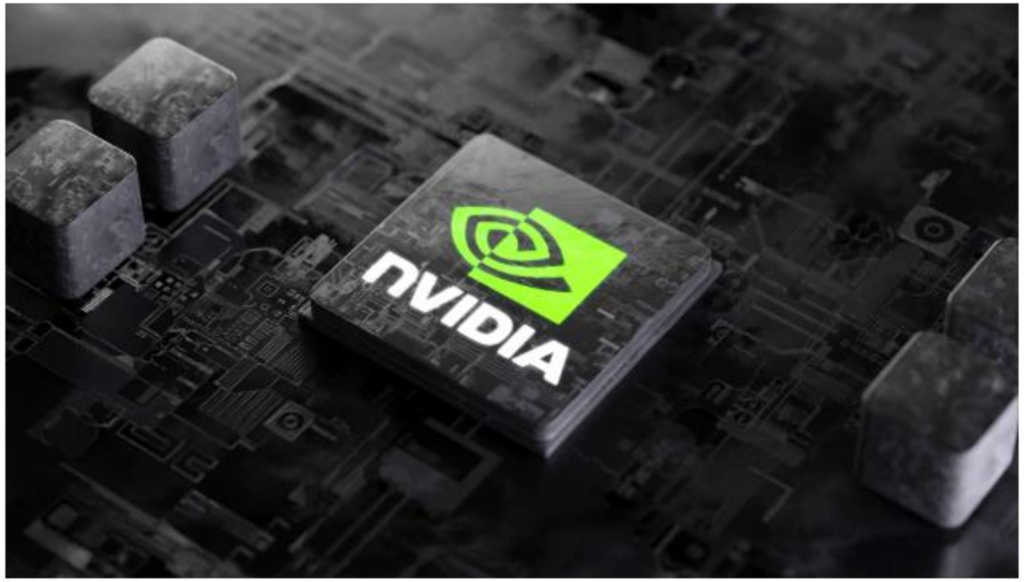

Nvidia have unveiled a new updated AI processor, helping the chip's capacity and speed, in an effort to cement AI dominance for the company.

Coined the Grace Hopper Superchip, a combination graphics chip and processor, is set to get a boost from new memory types, relying on high bandwidth memory 3, or HBM3e, now able to access information at 5TB a second.

Known as GH200, this chip is set to begin production in Q2 2024, part of a new lineup of hardware and software announced at a computer-graphics expo.

Nvidia has an early lead in the AI accelerators markets, with chips excelling at crunching data while developing AI software, helping the company's valuation pass $1 trillion in 2023, becoming the world's most valuable chipmaker.

https://spectrum.ieee.org/homomorphic-encryption

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FEATURE SEMICONDUCTORS

CHIPS TO COMPUTE WITH ENCRYPTED DATA ARE COMING

Fully homomorphic encryption could make data unhackable

"It's a new chapter in the history of computing."

—RO CAMMAROTA, INTEL

Computing with FHE means doing transforms, addition, and multiplication on "a very long list of numbers, and each number in itself is very large," explains Rohloff. Computing with numbers that might require more than a hundred bits to describe is not something today's CPUs and GPUs are inherently good at. If anything, GPUs have been going in the opposite direction, focusing on less precise math done using smaller and smaller floating-point numbers. The FHE accelerator chips, by contrast, can stream huge volumes of data through hardware that does integer math on numbers that are thousands of bits long to accommodate encryption's precision needs.

Each accelerator has its own way of dealing with these streams of huge numbers. But they're all after the same goal—making FHE as fast as today's unencrypted computing.

https://pressnewsagency.org/how-nvidia-grew-to-a-trillion-dollar-firm-fueling-the-ai-revolution/

NVIDIA Chips

Business

How Nvidia grew to a trillion-dollar firm fueling the AI revolution

By **PressNewsAgency** February 24, 2024

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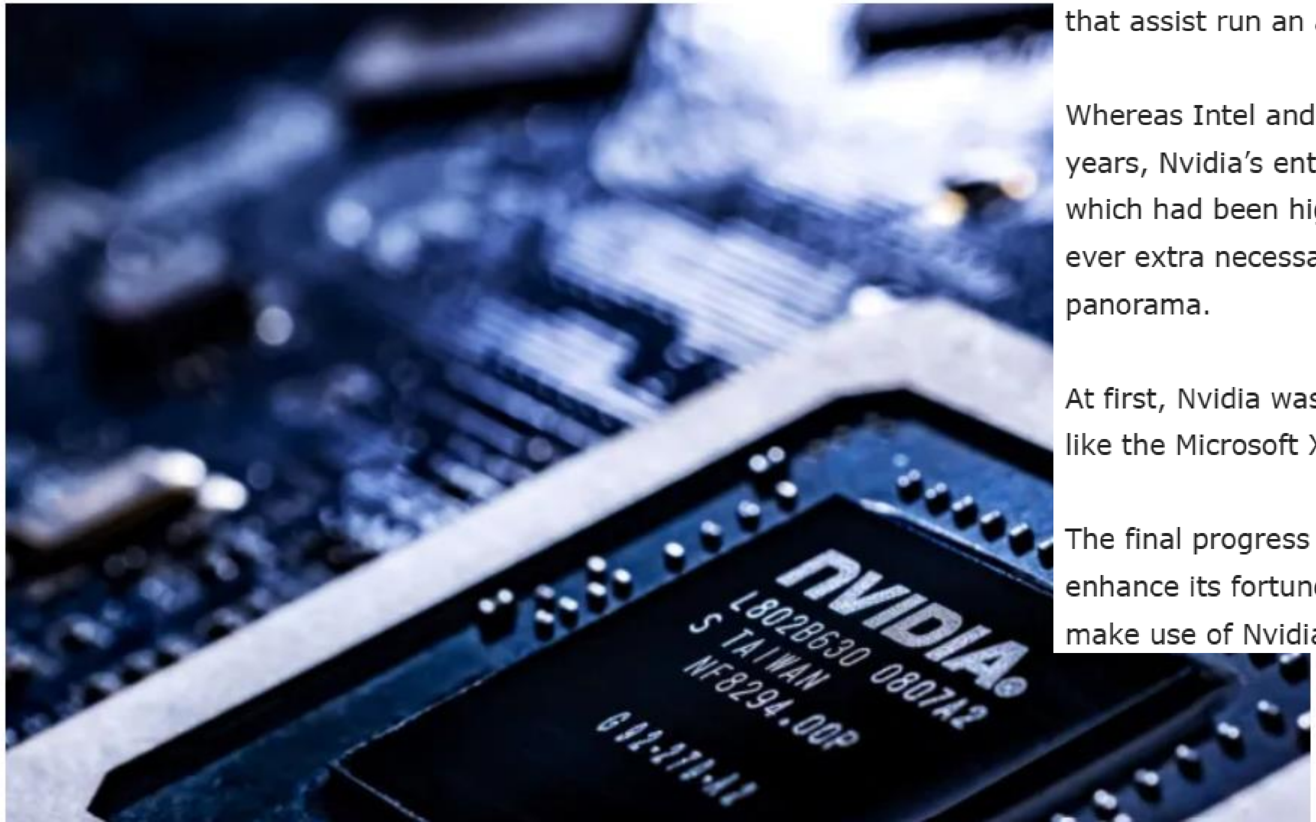


The enterprise world is more and more banking on synthetic intelligence to be the following huge factor, and has discovered itself turning to 1 maker of pc chips particularly — Nvidia — to energy the revolution.

Since 1993, the Santa Clara, California-based firm has been designing programmable chips that assist run an array of consumer-facing purposes.

Whereas Intel and Superior Micro Gadgets had dominated the U.S. chip sector for many years, Nvidia's entry signaled the arrival of refined graphics processing models (GPUs), which had been higher in a position to render photos. That functionality grew to become ever extra necessary as high-quality video more and more dominated the tech and media panorama.

At first, Nvidia was most related to offering GPU processors for online game consoles just like the Microsoft Xbox and Sony PlayStation.

The final progress of Silicon Valley through the 2010s prompted Nvidia to diversify and enhance its fortunes. For instance, in 2014, Nvidia and Google introduced a partnership to make use of Nvidia chips in Google Chromebooks.

https://www.livescience.com/technology/computing/scientists-create-worlds-1st-chip-that-can-protect-data-in-the-age-of-quantum-computing-attacks

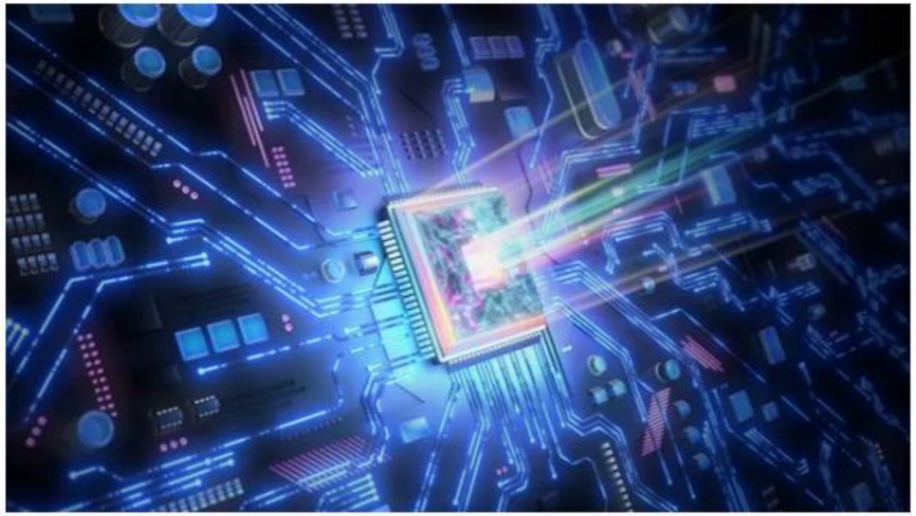
Technology > Computing

Scientists create world's 1st chip that can protect data in the age of quantum computing attacks

News By Peter Ray Allison published February 25, 2025

Scientists in Switzerland have developed a new method to improve internet security against quantum computing attacks, using quantum-resistant encryption and a new type of hardware.

Comments (0)



The QS7001 system combines two quantum-resistant encryption protocols to reduce the window of opportunity for attackers. (Image credit: Yuichiro Chino/Getty Images)

Engineers have demonstrated a new communications system designed to protect telecommunications against quantum computing attacks.

https://hailo.ai/products/ai-vision-processors/hailo-15-ai-vision-processor/#hailo15-overview

HAILO

Unprecedented AI performance in a camera power envelope

Hailo-15 is a series of AI vision processors for smart cameras. The Hailo-15 VPU System-on-a-Chip (SoC) combines Hailo's patented and field-proven AI inferencing capabilities with advanced computer vision engines, generating premium image quality and advanced video analytics. The unprecedented AI capacity of our vision processing unit can be used for both AI-powered image enhancement and processing of multiple complex deep learning AI applications at full scale and at superior efficiency.

LUMS

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Article | Published: 29 May 2024

A vision chip with complementary pathways for open-world sensing

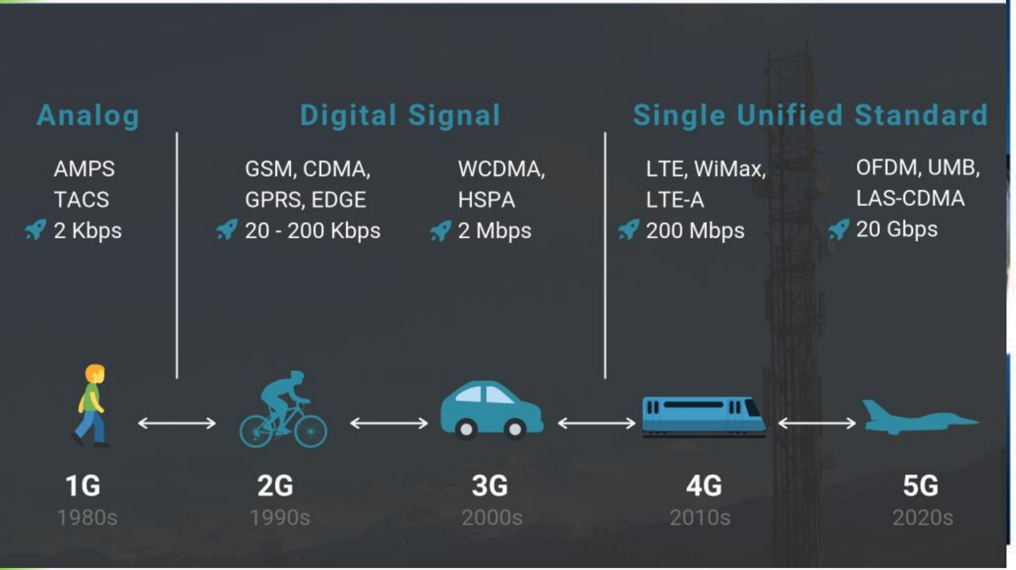
Zheyu Yang, Taoyi Wang, Yihan Lin, Yuguo Chen, Hui Zeng, Jing Pei, Jiazheng Wang, Xue Liu, Yichun Zhou, Jianqiang Zhang, Xin Wang, Xinhao Lv, Rong Zhao & Luping Shi

Nature **629**, 1027–1033 (2024) | Cite this article

Matters Arising to this article was published on 04 June 2025

Abstract

Image sensors face substantial challenges when dealing with dynamic, diverse and unpredictable scenes in open-world applications. However, the development of image sensors towards high speed, high resolution, large dynamic range and high precision is limited by power and bandwidth. Here we present a complementary sensing paradigm inspired by the human visual system that involves parsing visual information into primitive-based representations and assembling these primitives to form two complementary vision pathways: a cognition-oriented pathway for accurate cognition and an action-oriented pathway for rapid response. To realize this paradigm, a vision chip called Tianmouc is developed, incorporating a hybrid pixel array and a parallel-and-heterogeneous readout architecture. Leveraging the characteristics of the complementary vision pathway, Tianmouc



eletimes.com/worlds-first-5g-communication-module-for-automotive-using-qualcomm-chips-revealed

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News & Analysis Technology Design White Papers Manufacturing Priority Markets

Home › New Products › World's first 5G communication module for automotive using Qualcomm chips revealed

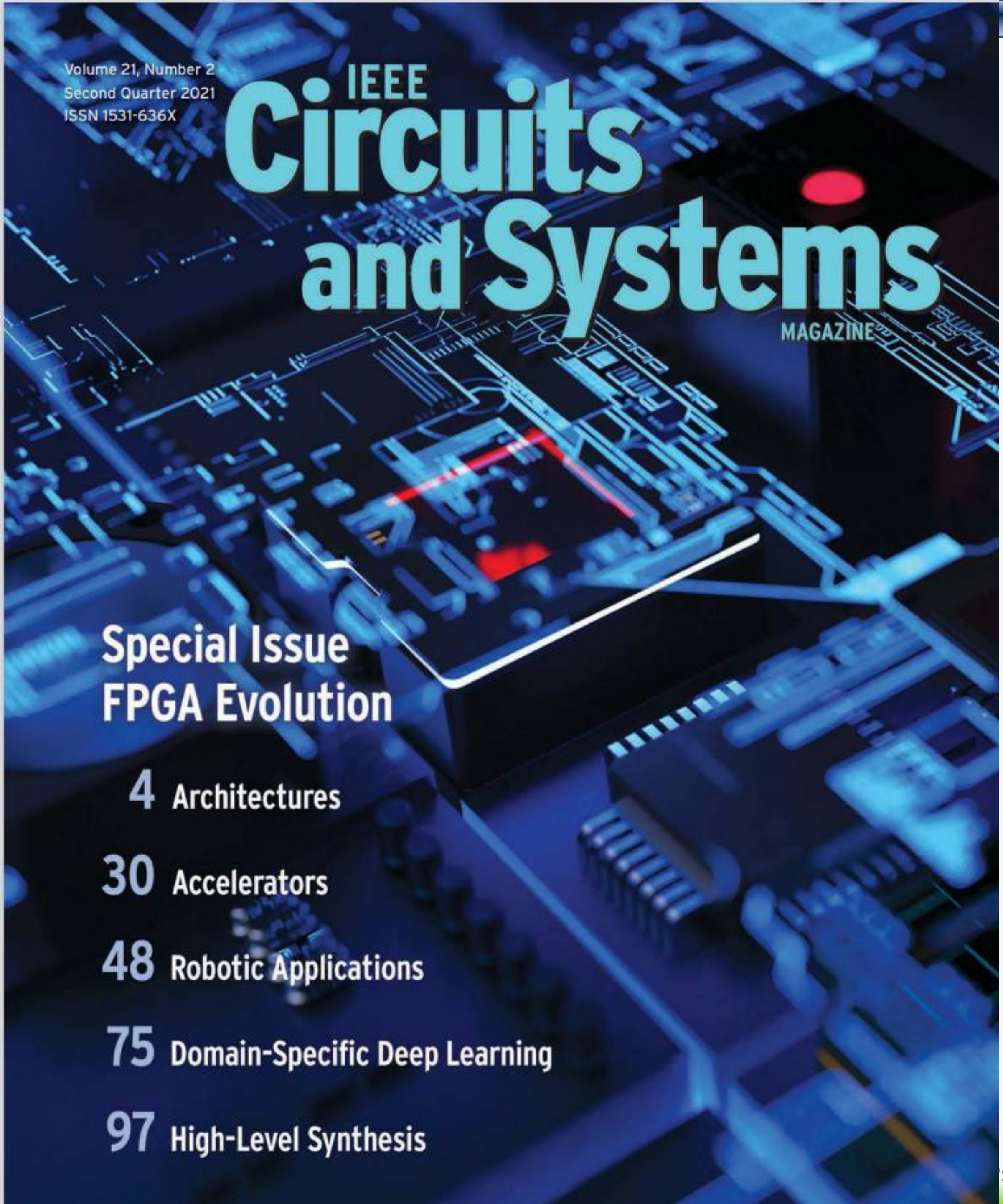
World's first 5G communication module for automotive using Qualcomm chips revealed

By **ELE Times** - October 17, 2019

LG Innotek announced it has developed a communication module for automotive based on 5G (5th Generation) Qualcomm chip. The company is the first company to develop a 5G communication module for automotive using Qualcomm chips that can be applied to vehicles.

5G communication module for automotive is a component that uses the 5G mobile telecommunication technology to enable transmission of data between a vehicle and a cellular base station and wireless network connection. This module combines a communication chip, memory, and RF (radio frequency) circuit and is generally mounted on the inside a vehicle or in a vehicle communication device on the roof.

This module makes possible sharing of real-time traffic information, precise location measurement, V2X (Vehicle-to-Everything: communication between a vehicle and objects), and transmission of larger amounts



IEEE Circuits and Systems Magazine – Special Issue on FPGA - 2021

FPGA is an Implementation Platform
To facilitate:
Rapid Design
Prototyping New Products
Design Re-Use
Easier Testing and Verification
Predictable Performance
…. Any many other useful features

TI Chips

https://www.ti.com/applications/automotive/adas/products.html

Advanced driver assistance systems (ADAS)

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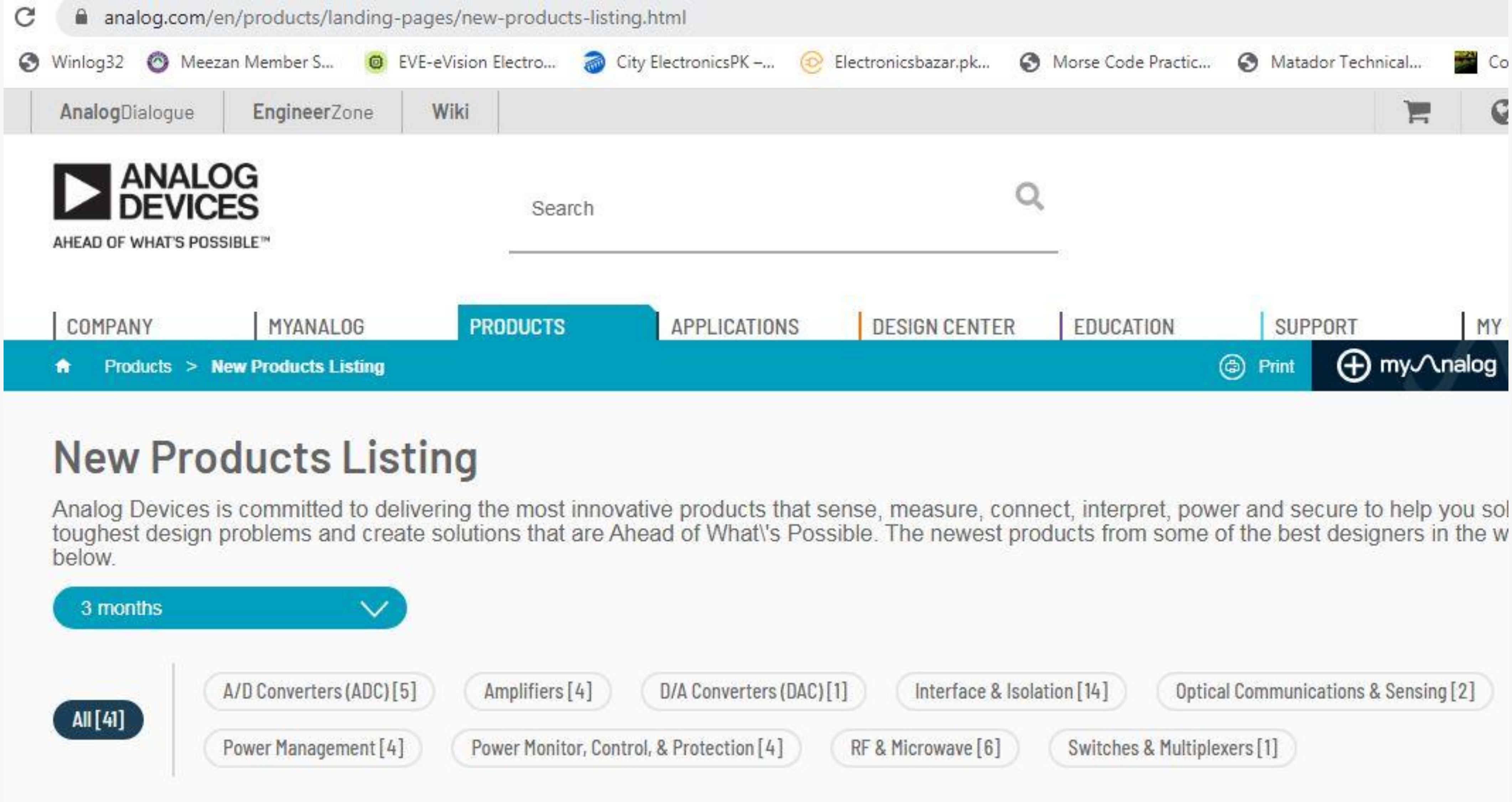
nber	Images	Description	Category	Subcategory	Operating temperature range (°C)
SPM0G3518-Q1 – NEW ta sheet: PDF | HTML		Automotive 80MHz ARM® Cortex®-M0+ MCU with 256kB flash, 128kB SRAM, 2 CAN, 2 ADC, DAC, COMP	Microcontrollers (MCUs) & processors	Arm Cortex-M0+ MCUs	-40 to 125
79758B-Q1 – NEW ta sheet: PDF | HTML		Automotive, 18-S ASIL-D compliant precise battery monitor and balancer with current sense	Battery management ICs	Battery monitors & balancers	-40 to 125
AN1476-Q1 – NEW ta sheet: PDF | HTML		Automotive dual signal improvement capable CAN FD transceiver with standby and 1.8V IO support	Interface	CAN transceivers	-40 to 150
IN4029A-Q1 – NEW ta sheet: PDF		Local interconnect network (LIN) transceiver with dominant state timeout and ±70V fault protection	Interface	LIN transceivers	-40 to 125
74AC132-Q1 – NEW ta sheet: PDF | HTML		Automotive four-channel 2-input 2V-to-6V NAND gate	Logic & voltage translation	NAND gates	to



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Amplifiers [4]
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Power Monitor, Control, & Protection [4]
RF & Microwave [6]
Switches & Multiplexers [1]

https://www.hackster.io/news/the-open-source-ztachip-is-a-risc-v-accelerator-for-edge-ai-and-computer-vision-applications-3539b7fba227

The Open Source Ztachip Is a RISC-V Accelerator for Edge AI and Computer Vision Applications

Designed to outperform even RISC-V chips with the recently-ratified vector extensions, ztachip can boost performance by up to 50x.

Gareth Halfacree
3 years ago • AI & Machine Learning / FPGAs

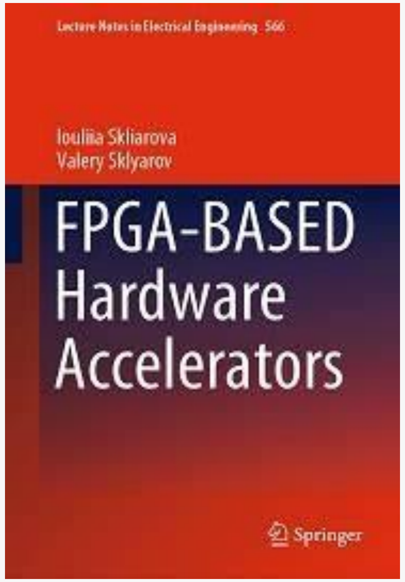
Follow

Embedded developer Vuong Nguyen has released an open source RISC-V accelerator designed to boost the performance of edge AI and computer vision tasks up to 50 times — and you can try it out yourself by loading it onto a field-programmable gate array (FPGA).

Examples of FPGA Based Custom Digital Systems

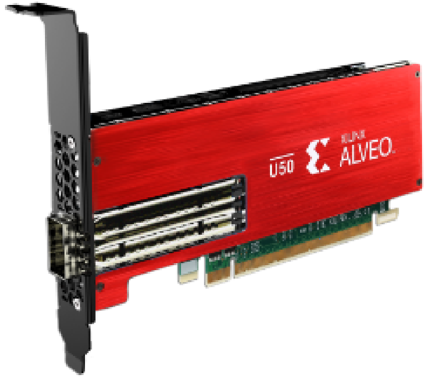
Using FPGA as an Implementation Platform for Complex Digital Circuits

Big Data Hardware Accelerator



/ Boards / Alveo / Alveo U50 Data Center Accelerator Card

Alveo U50 Data Center Accelerator Card



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Product Description

The AMD Alveo™ U50 Data Center accelerator cards provide optimized acceleration for workloads in financial computing, machine learning, computational storage, and data search and analytics. Built on AMD UltraScale+ architecture and packaged up in an efficient 75-watt, small form factor, and armed with 100 Gbps networking I/O, PCIe Gen4, and HBM, Alveo U50 is designed for deployment in any server.

Alveo accelerator cards are adaptable to changing acceleration requirements and algorithm standards, capable of accelerating any workload without changing hardware, and reduce overall cost of ownership.

Enabling Alveo accelerator cards is an ecosystem of AMD and partner applications for common Data Center workloads. For custom solutions, Application Developer Tool Suite (Vitis™ environment) and Machine Learning Suite from AMD provide the frameworks for developers to bring differentiated applications to market.

FPGA in Big Data

FPGA Acceleration for Big Data Analytics: Challenges and Opportunities

Joost Hoozemans, Johan Peltenburg, Fabian Nonnenmacher, Ákos Hadnagy, Zaid Al-Ars, and H. Peter Hofstee

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Abstract

The big data revolution has ushered an era with ever increasing volumes and complexity of data requiring ever faster computational analysis. During this very same era, CPU performance growth has been stagnating, pushing the industry to either scale their computation horizontally using multiple nodes in datacenters, or to scale vertically using heterogeneous components to reduce compute time. However, networking and storage continue to provide both higher throughput and lower latency, which allows for leveraging heterogeneous components, deployed in

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Date of current version: 24 May 2021

30 IEEE CIRCUITS AND SYSTEMS MAGAZINE 1531-636X/21©2021IEEE SECOND QUARTER 2021

FPGA and the Cloud

IEEE TRANSACTIONS ON CLOUD COMPUTING, VOL. 10, NO. 2, APRIL-JUNE 2022

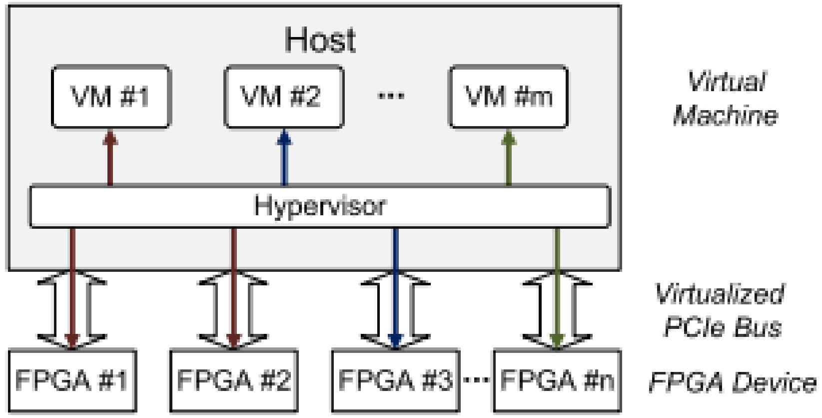

Fig. 4. Architecture of cloud FPGA instance.

1344 IEEE TRANSACTIONS ON CLOUD COMPUTING, VOL. 10, NO. 2, APRIL-JUNE 2022

When FPGA Meets Cloud: A First Look at Performance

Xiuxiu Wang, Yipei Niu, Fangming Liu, *Senior Member, IEEE*, and Zichen Xu, *Member, IEEE*

Abstract—Cloud service providers promote their new field programmable gate array (FPGA) infrastructure as a service (IaaS) as the new era of cloud product. This FPGA IaaS wraps virtualized compute resources with FPGA boards, e.g., Amazon AWS F1, and reserves acceleration capability for specific applications. Though this acceleration technique sounds promising, questions like real world performance, best-fit scenarios, portability, etc., still need further clarification. In this article, we present one of the first few empirical studies that take a close look at FPGA clouds from the tenants' perspective. We have conducted measurement studies on Amazon AWS, Alibaba, and Huawei clouds for over one year. The experimental results show that: (1) Tenants experience severe performance-cost imbalance on FPGA IaaS platforms; (2) The inter-communication performance in FPGA clouds is tightly constrained by hardware drivers, e.g., small optimization of DMA drivers for PCIe can harvest significant performance gain; (3) The virtualized FPGA clouds are far from mature, e.g., small-sized jobs can greatly degrade the performance of FPGA clouds due to underutilized PCIe bandwidth. Our study not only provides useful hints to help tenants with FPGA service selection, but also sheds some lights for cloud providers to improve the performance of FPGA clouds.

Index Terms—FPGA cloud, FPGA acceleration, virtualization, performance measurement

FPGA Tools on Amazon Web Services AWS

FPGA in CNN – YOLO Example

IEEE Access
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Digital Object Identifier 10.1109/ACCESS.2020.3004198

Sparse-YOLO: Hardware/Software Co-Design of an FPGA Accelerator for YOLOv2

ZIXIAO WANG[1,2], KE XU[1,2], SHUAIXIAO WU[1,2], LI LIU[3], LINGZHI LIU[3], (Senior Member, IEEE), AND DONG WANG[1,2], (Member, IEEE)

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Corresponding author: Dong Wang (wangdong@bjtu.edu.cn)

This work was supported in part by the National Key Research and Development Program of China under Grant 2019YFB2204200, in part by the Beijing Natural Science Foundation under Grant 4202063, and in part by BJTU-Kuaishou Research Grant.

ABSTRACT Convolutional neural network (CNN) based object detection algorithms are becoming dominant in many application fields due to their superior accuracy advantage over traditional schemes. Among them, You Look Only Once (YOLO) is one of the most popular detection frameworks that show best trade-offs between speed and accuracy. However, due to the intrinsic high computational workload of CNN, it is still challenging when targeting high-throughput processing with low cost in energy consumption. In this paper, we propose a hardware/software (HW/SW) co-design methodology targeting CPU+FPGA-based heterogeneous platforms. Firstly, we extend a novel sparse convolution algorithm to the YOLOv2 framework, and then develop a resource-efficient FPGA accelerator architecture based on asynchronously executed parallel convolution cores. Secondly, algorithm-level optimization schemes, including hardware-aware neural network pruning, clustering and quantization are introduced, which successfully save the computational workload of the YOLOv2 algorithm by 7 times. Finally, an end-to-end design space exploration flow for FPGA-based accelerator design is presented and two HW/SW partition strategies are studied and implemented. Experimental results show that our design can achieve a peak throughput of 2.13 TOPS (72.5 fps) on an Intel Arria-10 GX1150 FPGA under the working frequency of 211 MHz, while the detection accuracy is 74.45 on the PASCAL VOC2007 dataset.

INDEX TERMS Convolutional neural networks, fine-grained pruning, field programmable gate arrays, object detection, YOLO.

FPGA in Electrical Systems

System-on-Chip FPGA Devices for Complex Electrical Energy Systems Control

ERIC MONMASSON, MICKAËL HILAIRET, GIOVANNI SPAGNUOLO, and MARCIAN N. CIRSTEA

Digital Object Identifier 10.1109/MIE.2021.3052179
Date of current version: 19 February 2021

Digital electronics has become a standard for controlling electrical systems. This is due to the constant improvement of the digital devices, whether in terms of density, performance, flexibility of use, or cost reduction [1]. This article looks into system-on-chip (SoC) field-programmable gate array (FPGA) for controlling complex electrical energy systems. These devices encompass multicore floating-point microprocessors embedded with standard peripherals and an FPGA fabric that allows the design of custom peripherals and specific hardware (HW) accelerators. Thus, SoC FPGA devices

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FPGA in Biomedical

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Efficient CNN Architecture on FPGA Using High Level Module for Healthcare Devices

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Corresponding author: Ahmed K. Jameil (2006957@brunel.ac.uk)

This work was supported by Brunel University London.

ABSTRACT Modern wearable healthcare devices require new technologies with resource efficiency in terms of high performance, low energy consumption and diagnostic accuracy. In the field of artificial intelligence, the convolutional neural network (CNN) has performed as an effective algorithm. Field-programmable gate arrays (FPGAs) have been extensively utilised to construct hardware accelerators for CNNs. This paper suggests using an accelerator to create a specific 1-D CNN to classify the electrogram (ExG). ExGs used here include electrocardiogram, electroencephalogram and electromyography. The pipelined structure is designed with a register in the middle to facilitate easy data transfer. A 1-D CNN using an accelerator to categorise ExG signals implemented on Xilinx Zynq xc7z045 platform outperforms FPGA peer applications on the same platform by 1.14× in terms of speed. In addition, the 1-D CNN proposed accelerator operates very efficiently due to the use of a tristate buffer in the multiplexer and the substitution of the shift for the multiplier, resulting in a resource-efficient accelerator with 161 GOP/s/W energy efficiency and 28 GOP/s/KLUT, an improvement of 1.67 over the previous model. Finally, the performance of the accelerator applied to a Xilinx Zynq xc7z045 FPGA operating at 442.948MHz was calculated, achieving 1.145 TFLOP/s.

INDEX TERMS Electrogram classification, convolutional neural network (CNN), FPGA, healthcare monitoring, hardware accelerator.

FPGA in Robotics

A Survey of FPGA-Based Robotic Computing

Zishen Wan,* Bo Yu,* Thomas Yuang Li, Jie Tang, Yuhao Zhu,
Yu Wang, Arijit Raychowdhury, and Shaoshan Liu

Abstract

Recent researches on robotics have shown significant improvement, spanning from algorithms, mechanics to hardware architectures. Robotics, including manipulators, legged robots, drones, and autonomous vehicles, are now widely applied in diverse scenarios. However, the high computation and data complexity of robotic algorithms pose great challenges to its applications. On the one hand, CPU platform is flexible to handle multiple robotic tasks. GPU platform has higher computational capacities and easy-to-use development frameworks, so they have been widely adopted in several applications. On the other hand, FPGA-based robotic accelerators are becoming increasingly competitive alternatives, especially in latency-critical and power-limited scenarios. With specialized designed hardware logic and algorithm kernels, FPGA-based accelerators can surpass CPU and GPU in performance and energy efficiency. In this paper, we give an overview of previous work on FPGA-based robotic accelerators covering different stages of the robotic system pipeline. An analysis of software and hardware optimization techniques and main technical issues is presented, along with some commercial and space applications, to serve as a guide for future work.

I. Introduction

Over the last decade, we have seen significant progress in the development of robotics, spanning from algorithms, mechanics to hardware platforms. Various robotic systems, like manipulators, legged robots, unmanned aerial vehicles, self-driving cars

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* These authors contributed equally to this work.
Corresponding author: Shaoshan Liu (email: shaoshan.liu@perceptin.io).

Special Issue

Advances in Field-Programmable Gate Array (FPGA)-Based Reconfigurable Systems

Message from the Guest Editor

FPGAs enable a designer to create custom, application-specific architectures to exploit algorithmic parallelism. Moreover, FPGAs are typically more energy efficient than a general-purpose processor, enabling a designer to achieve massive computational speedup on an application while consuming less energy. The following points are problematic in the development of reconfigurable systems: deficiency of high-level synthesis design languages and tools, operating systems for reconfigurable systems, debugging and verification tools, partial reconfiguration techniques, and design techniques for self-aware, secure, reliable, and fault-tolerant reconfigurable systems. Its topics are not limited by the keywords listed below:

- reconfigurable systems
- field-programmable gate arrays (FPGAs)
- reconfigurable computing
- reconfigurable accelerators
- reconfigurable computing platforms and architectures
- reconfigurable processors
- FPGA partial reconfiguration
- methods and tools for reconfigurable computing
- fault-tolerant computing
- reconfigurable control systems

Guest Editor

Prof. Dr. Valery Salauyou
Faculty of Computer Science, Bialystok University of Technology, 15-351 Bialystok, Poland

Deadline for manuscript submissions

20 October 2025

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Special Issue

System-on-Chip (SoC) and Field-Programmable Gate Array (FPGA) Design

Message from the Guest Editor

The evolution of digital technology is firmly steered by key concepts such as field-programmable gate arrays (FPGAs) and the evolution of system-on-chip (SoC). These pivotal elements seamlessly integrate programmable logic, creating a potent synergy that manifests in spatial and temporal computing. While SoCs stand as a milestone in technological evolution, incorporating processors and reconfigurable logic areas, FPGAs emerge as fundamental devices for the rapid development of complex digital circuits. Together, SoCs and FPGAs create an ecosystem that not only accelerates the time to market but also opens the doors to a new computing paradigm, where flexibility and spatial–temporal optimization are at the forefront of digital innovation. The significance of these FPGA and SoC architectures extends beyond mere computing power, influencing the design of advanced devices in sectors such as artificial intelligence, the Internet of Things, and robotics, radically transforming how we interact with technology in our daily lives.

Guest Editor

Prof. Dr. Nicola Lusardi
Department of Electronics, Information and Bioengineering, Politecnico di Milano, 20133 Milano, Italy

Deadline for manuscript submissions

closed (15 January 2025)

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The CS community for FPGA

INTRODUCTION | FREE ACCESS

Introduction to the Special Section on FPGA 2023

Authors: Suhaib A. Fahmy, Jason D. Bakos | Authors Info & Claims

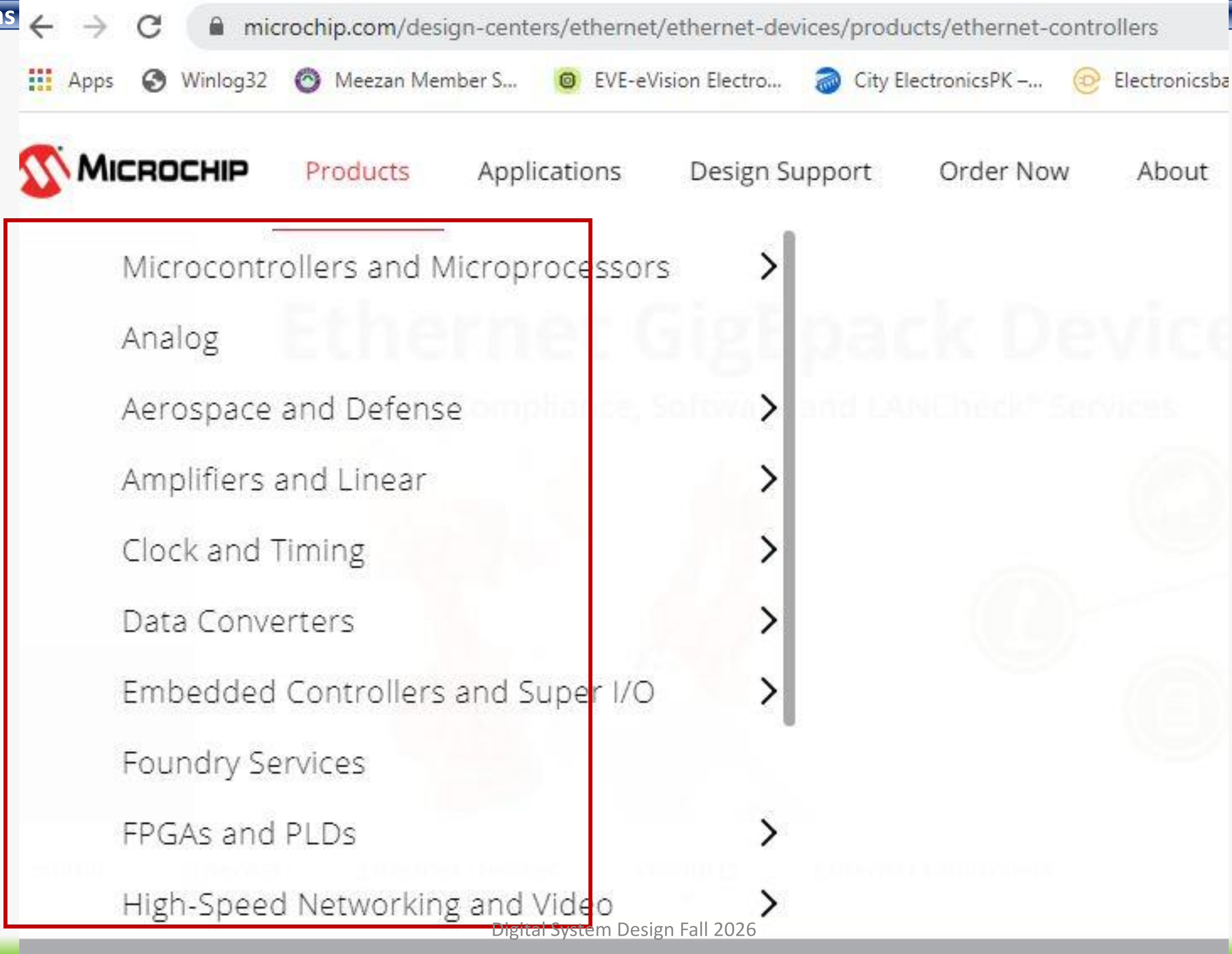
ACM Transactions on Reconfigurable Technology and Systems, Volume 17, Issue 3 • Article No.: 43, Pages 1 - 2
https://doi.org/10.1145/3695841

Published: 30 September 2024 Publication History

The ACM International Symposium on **Field-Programmable Gate Arrays (FPGA)** is one of the four premier venues for presenting and discussing advances in FPGA technology, tools, and applications. The 31st meeting was held in Monterey, CA, in February 2023 at a time when FPGA technology has become more crucial than ever for its potential to deliver the energy efficiency and low latency demanded by society's most important applications.

This special section comprises extended versions of eight selected papers presented at the conference. These papers describe novel work in solving a diverse set of challenges while capturing a comprehensive cross-section of the cutting-edge FPGA research in 2023.

e first article, "CSAIL2019 Crypto-Puzzle Solver Architecture," by Gribok et al., describes the orld's fastest solver for solving the CSAIL2019 time-lock puzzle: $2^{2^t} mod N$ for $t = 2^{56}$ and a 3,072-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Digital System Design for Everyone!!

Scale of Digital Systems

- Logic Gates
- Decoders, Mux, Glue Logic
- VGA Controller, DMA, Interfacing,
- Microcontrollers and Microprocessors
- Multicore Processors, DSP Chips, VR Chips
- ASIC – Application Specific Integrated Circuits
  FPGA – Field Programmable Gate Arrays

Small Scale Integration

Medium Scale Integration

Large Scale Integration

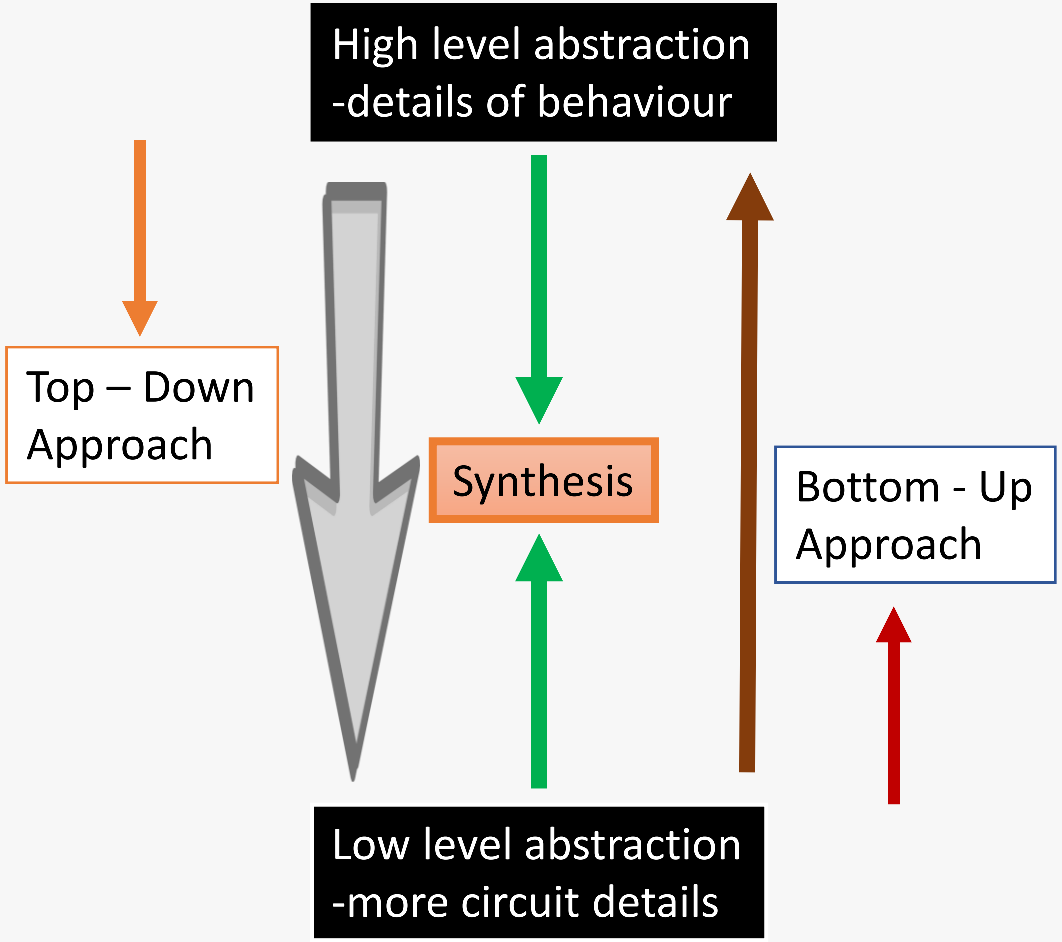
VLSI – Very Large-Scale Integration

- CMOS
- Nano-technology (6nm feature size)
- Reconfigurable Hardware

Handling the Complexity – Design Abstraction

Different hierarchy levels to describe digital systems:

- **Application Software System**
- **Operating System hardware drivers**
- **Computer Organization and Architecture**
- **Microarchitecture of CPU**
- **Digital System behavior**
- **Logic Gates and Circuits**
- **Discrete Components**
- **Semiconductor Devices**
- **Physics and Nano Technology**



Abstraction – Define circuit behaviour in English language

The HDL ↔ EDA ↔ FPGA connection

Idea

Design Specifications in schematics, behavior, pseudo-code, text

HDL to capture the idea and describe the design, both combinational and sequential

EDA – Simulate the design written in HDL (Verilog or VHDL languages)

EDA – Synthesis tools that map HDL code to physical logic gates

FPGA Chips – Reconfigurable hardware platform implementation of synthesis results

Rapid Prototoying of Complete projects

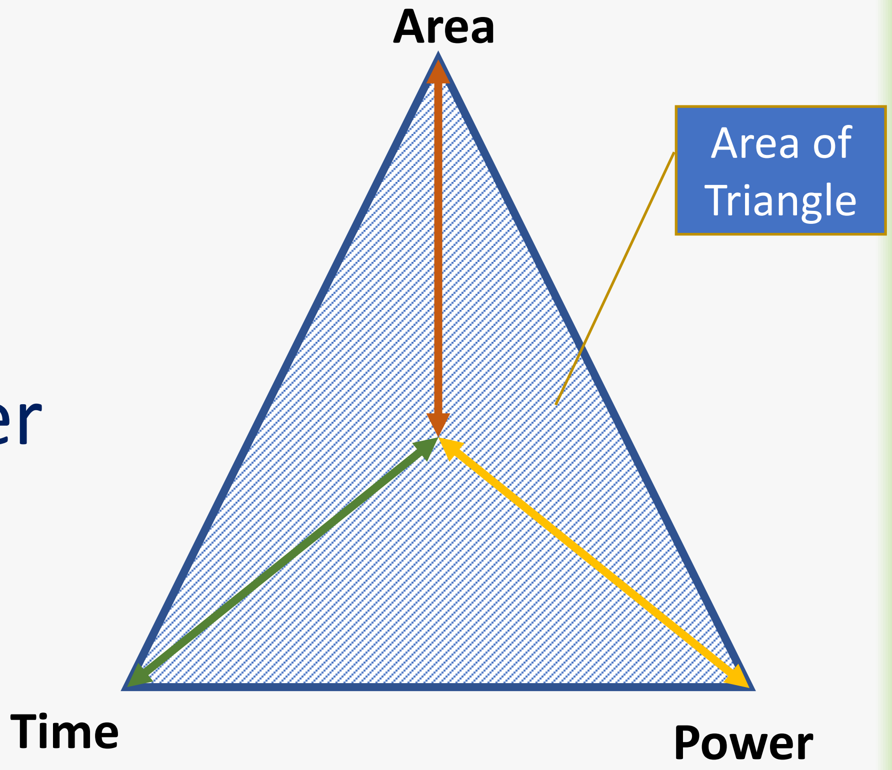
EDA = Electronic Design Automation
CAD = Computer Aided Design

Rapid Prototyping ?

Conclusion

- Digital Systems are Ubiquitous
- Important Aspects of Digital Systems
  - Design Capture, Complexity management
  - Simulation
  - Synthesis on Different Platforms
- Implementation Technologies
- Performance matrix – area, speed, power

Modern Design Considerations



Course Outline and Grading